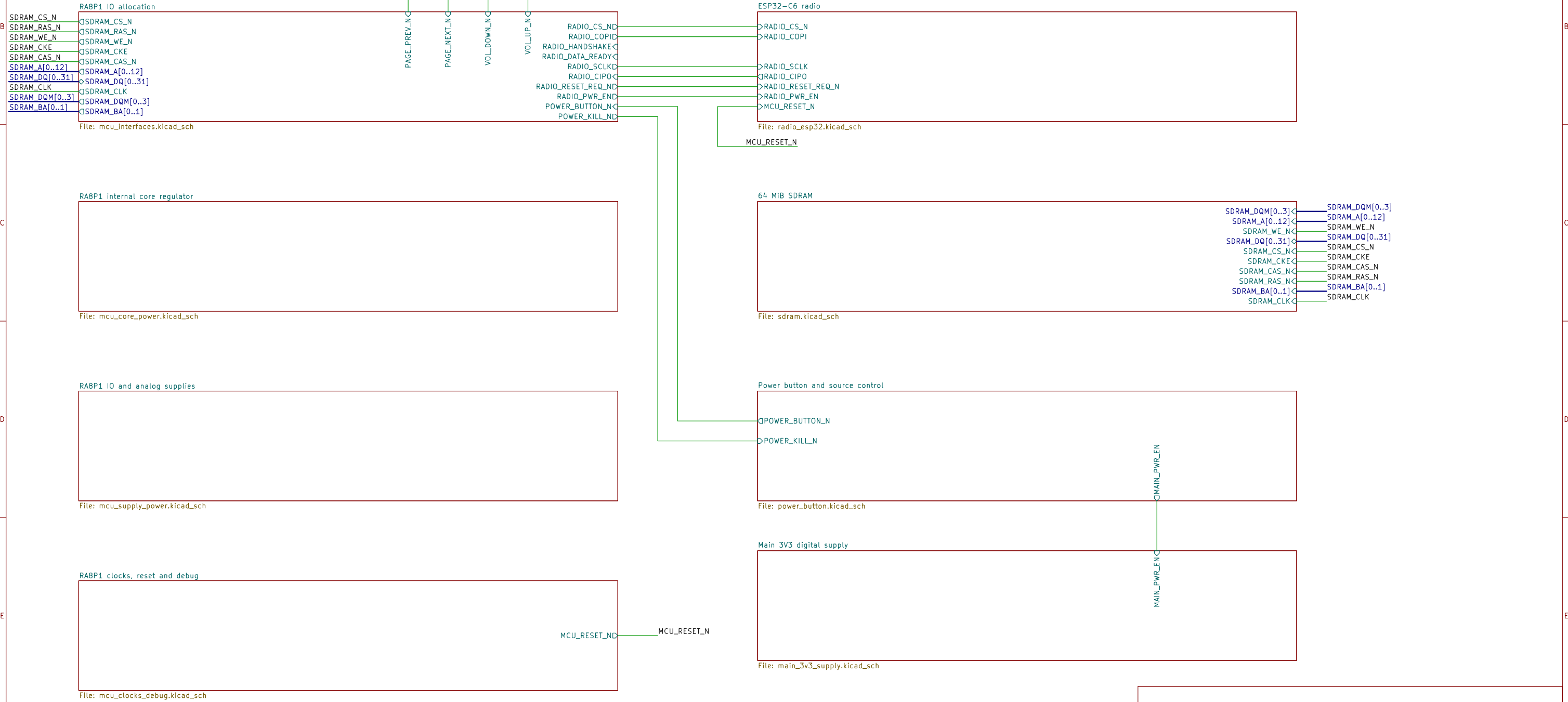


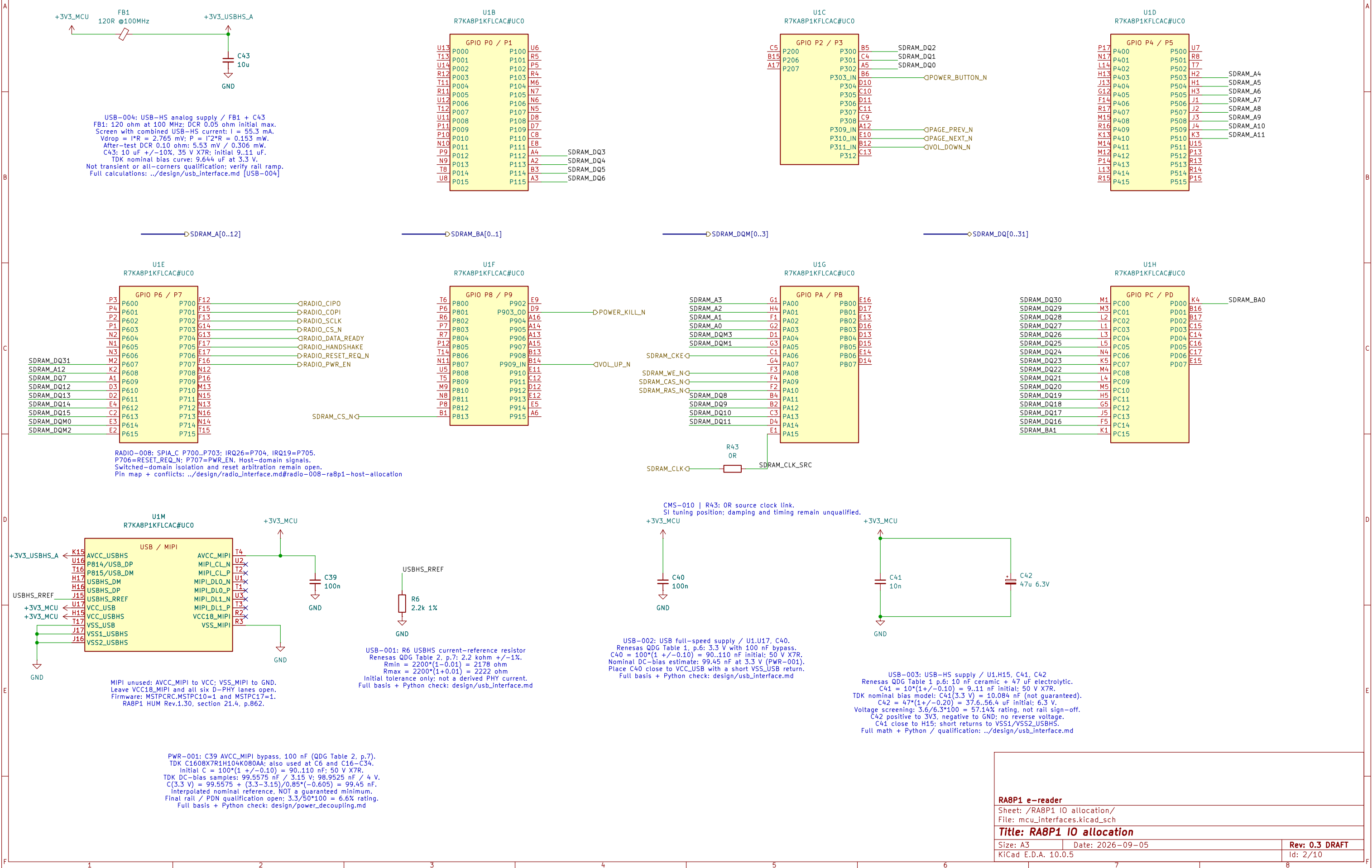
# RA8P1 E-READER

Development draft – not for fabrication.  
Requirements: design/ereader\_requirements.md  
Tracking: GitHub epic #821.

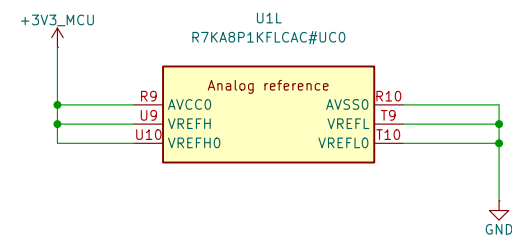
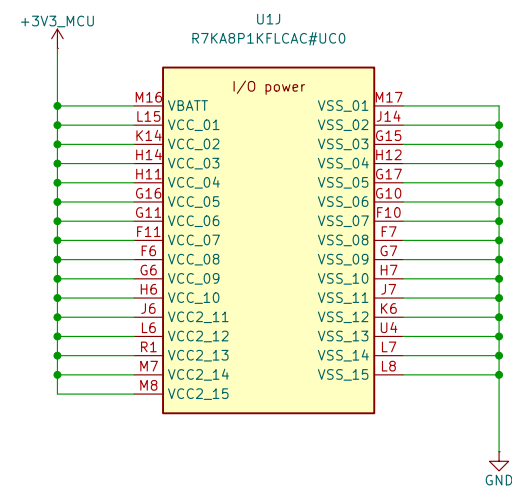


Global supplies: +3V3\_MCU and GND.  
MCU\_VCORE remains local to the processor core-power sheet.  
Circuit notes link to the supporting engineering calculations.  
Footprint and PCB qualification are deferred by scope.

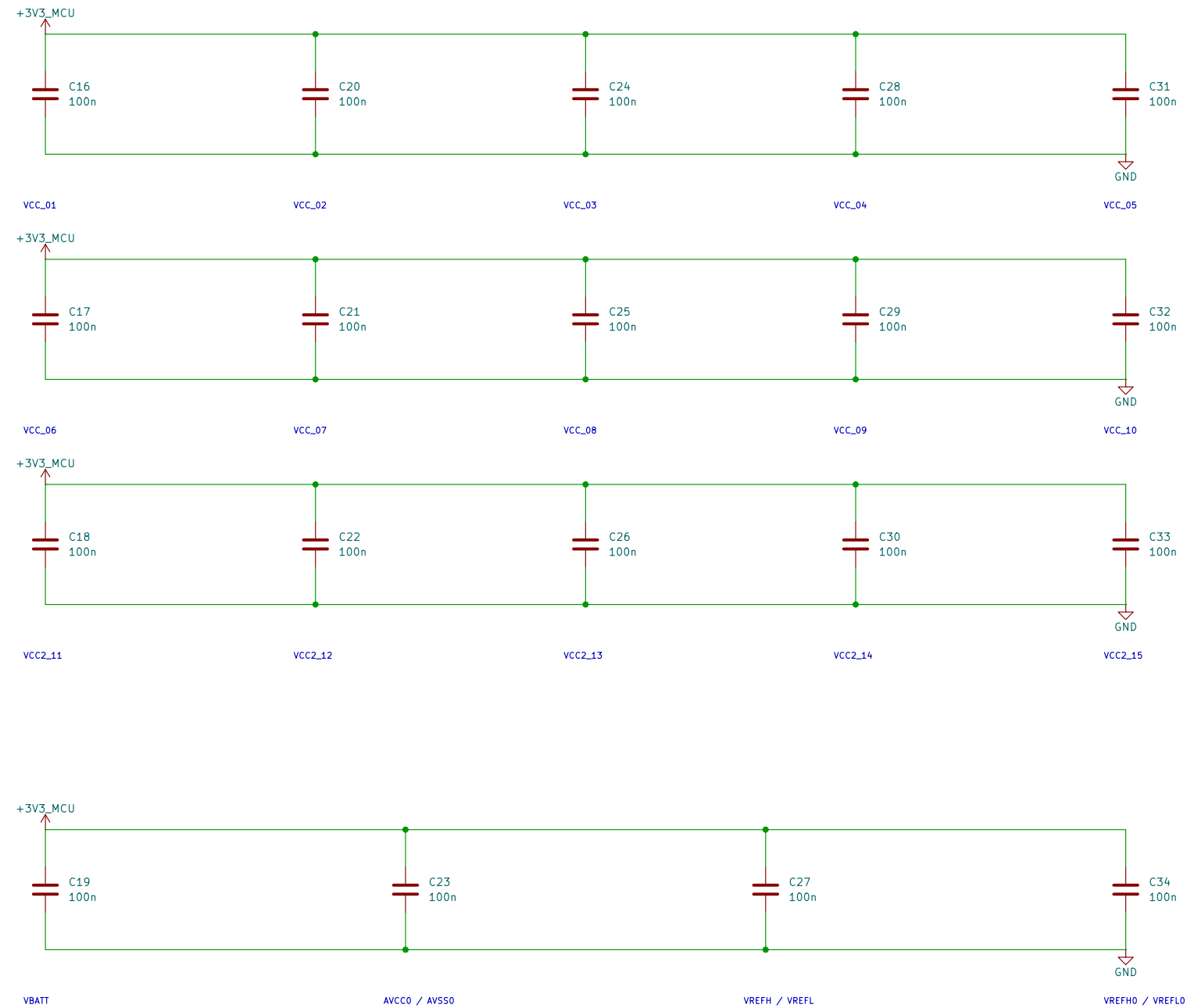
DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.



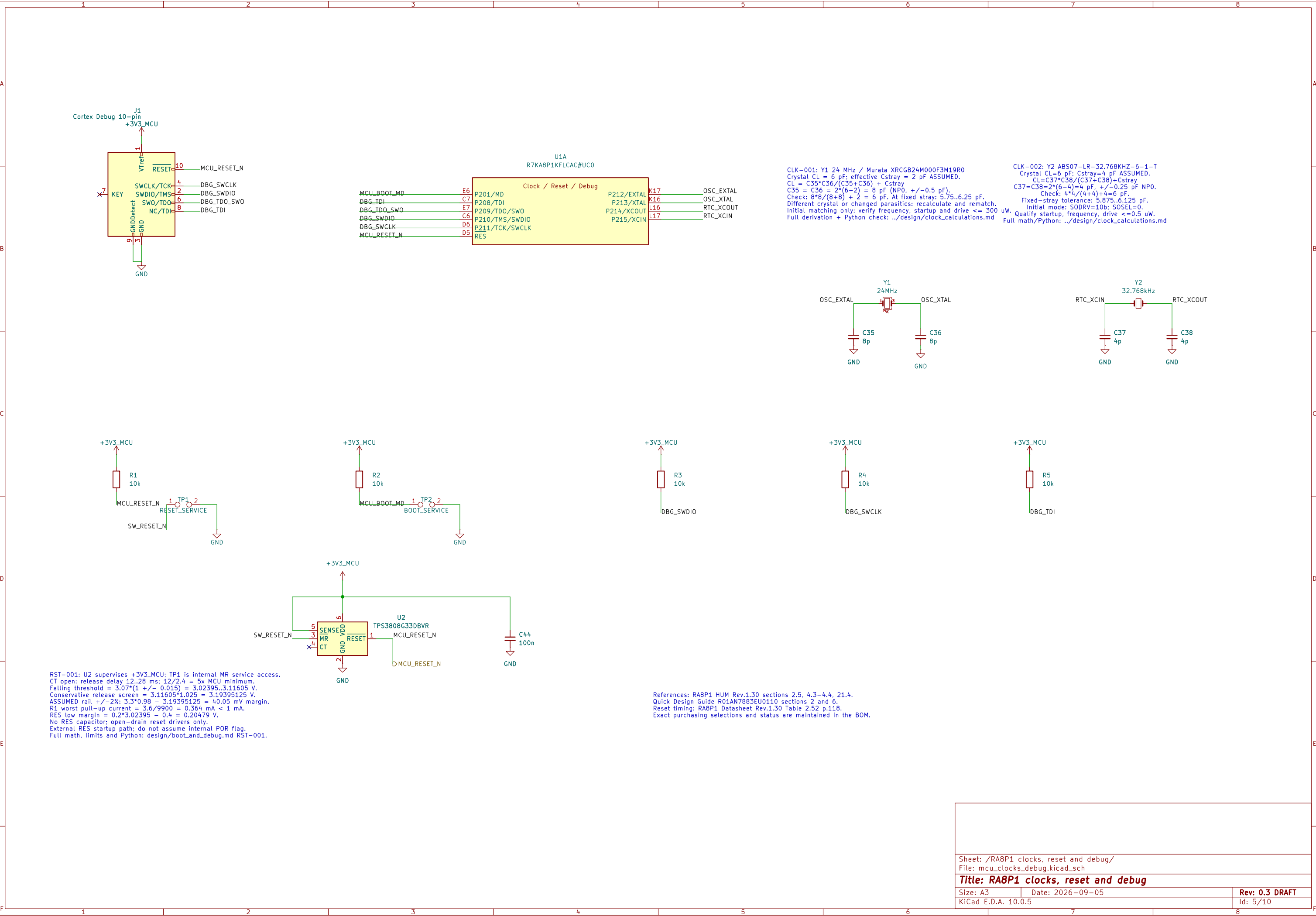




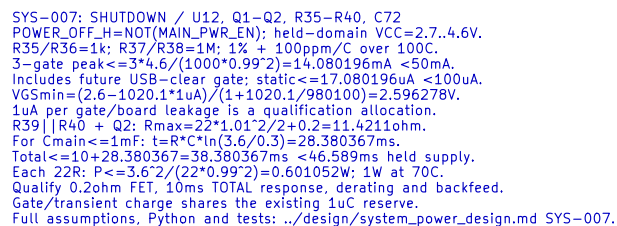
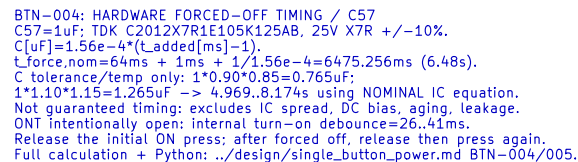
I/O DECOUPLING – 100 nF AT EACH NUMBERED VCC / VSS PAIR



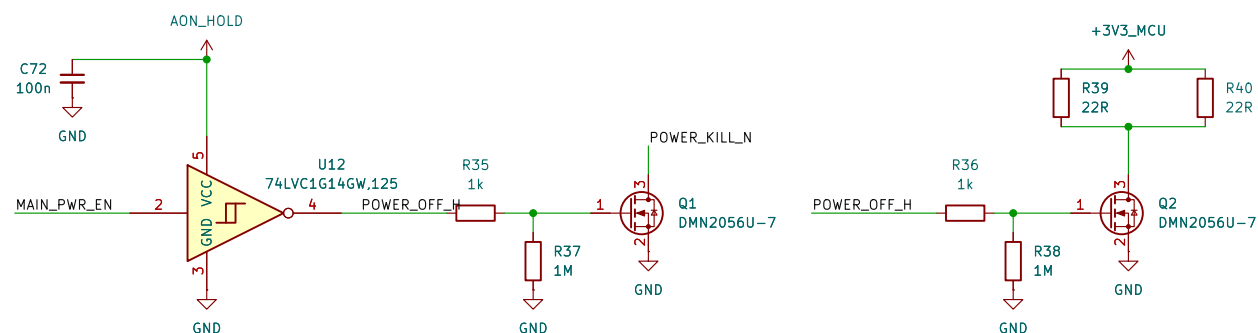
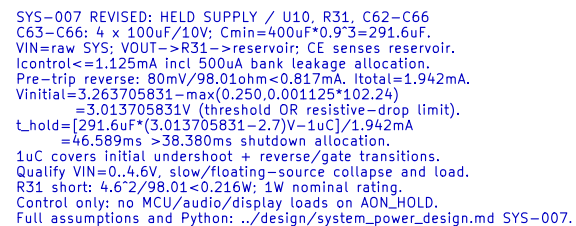
VCC and VCC2 use 3.3 V. All returns use one board ground plane. VBATT follows the main rail; no separate backup-cell supply. Analog references use AVCC0 / AVSS0. Place bypasses at named ball pairs. USB/MIP1 supplies remain on the interface sheet and are not yet wired. HUM Rev.1.30; Table 69.2 p.4042; unused references: section 21.4 pp.861–862. This is an electrical draft; capacitor MPN qualification remains open.





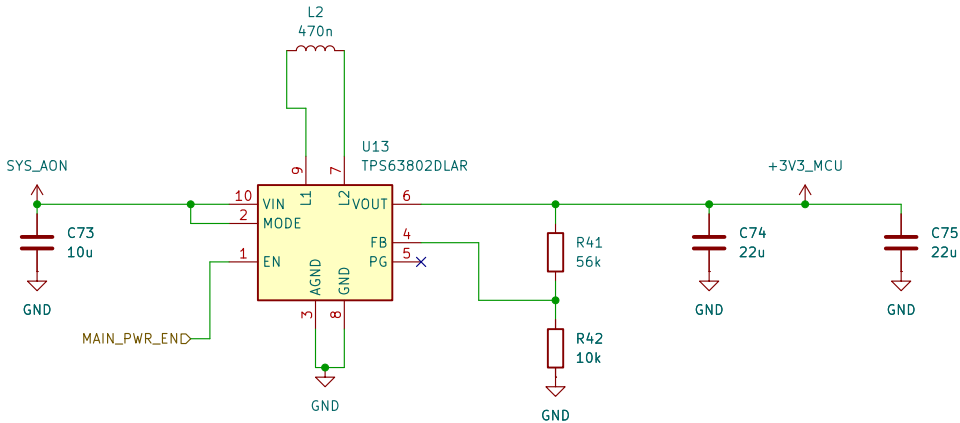


ERC supply path: U10 VOUT through R31.  
PWR\_FLAG declares that series-fed source only;  
raw SYS\_AON source remains unimplemented.





MAIN DIGITAL 3.3 V SUPPLY  
TPS63802 buck–boost; hardware–controlled enable; fixed PWM.  
Engineering draft: upstream source and corner qualification remain open.



PWR–002 / PWR–003: SETPOINT AND LOCAL COMPONENTS

U13 TPS63802DLAR; MODE=VIN, fixed PWM.  
R41=56k; R42=10k; each +/-0.1%, 25ppm/C.  
Vnom=0.500\*(1+56k/10k)=3.300V.  
Idivider=3.300/(56k+10k)=50uA.  
For 100C resistor–temperature change:  
fLo=0.999\*0.9975=0.9965025; fHi=1.001\*1.0025=1.0035025.  
Vmin=0.495\*(1+56k\*fLo/(10k\*fHi))–100nA\*56k\*fHi  
=3.242044V.  
Vmax=0.505\*(1+56k\*fHi/(10k\*fLo))+100nA\*56k\*fHi  
=3.358485V.  
Static screen only; ripple, load steps and drift still require validation.  
C73: CIN\_model=10u\*0.90\*0.85\*0.60=4.590uF >=4uF.  
C74+C75: COUT\_model=2\*22u\*0.90\*0.85\*0.60=20.196uF >=7uF.  
0.90=tolerance; 0.85=X7R temperature; 0.60=residual ALLOCATION.  
Qualify bias, aging, excitation and assembly; 60% is not guaranteed.  
L2: 0.47uH +/-20% gives 0.376..0.564uH INITIAL only.  
Required installed effective L=0.37..0.57uH; bias/reflow/hot tests OPEN.  
TI SLVSEU9D Table 7–1, sections 9.4/10.2; exact parts in native BOM.  
Full derivation, primary sources and executable Python:  
../design/power\_decoupling.md PWR–002/PWR–003.

PWR–003: LOAD, SOURCE AND STARTUP ACCEPTANCE LIMITS

Main continuous allocation: 0.750 MCU + 0.500 radio  
+ 0.250 SDRAM + 0.100 NOR + 0.050 logic = 1.650A.  
These are qualification allocations, not measured maximum currents.  
SD, camera, audio, display/touch/frontlight need separate SYS supplies.  
Pout=3.358485094\*1.65=5.541500W.  
VIN>=3.20V allocation; raw trip min=3.263705831V.  
Allowed raw–to–VIN loss=3.263705831–3.20=63.705831mV.  
At ASSUMED eta=85%: Iin=Pout/(eta\*3.20)=2.037316A.  
Ploss=Pout\*(1/eta–1)=0.977912W.  
Reference rise=Ploss\*81C/W=79.210859C; NOT enclosure prediction.  
Verify hot current capability and junction temperature <125C with margin.  
Cold subtotal, RADIO OFF: 1.330 DCDC + 0.400 other = 1.730A.  
Radio on would give 2.230A > nominal 2A; keep radio OFF through start/wake.  
Largest wake–reference subtotal=1.270+0.400=1.670A.  
DCDC figures are reference–only; non–DCDC start and extra charging excluded.  
Additional charging: I=Cadditional\*dV/dt; qualify the actual ramp.  
Total directly connected main–rail C <=1mF for SYS–007 discharge budget.  
No externally driven +3V3\_MCU: forced PWM can sink reverse current.  
PWR–002 static release margin: radio 14.922537mV; MCU 48.092861mV.  
Ripple, startup and transients must fit; no fabrication release.  
Full sources, derivation and Python: ../design/power\_decoupling.md PWR–003;  
source–loss shutdown: ../design/system\_power\_design.md SYS–007.

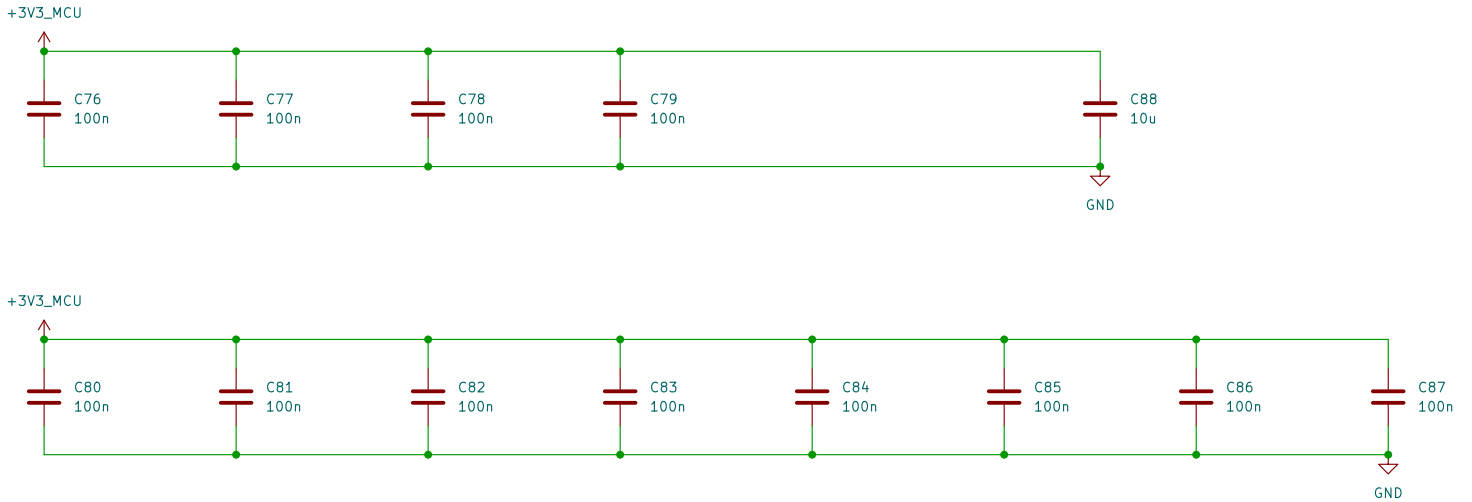
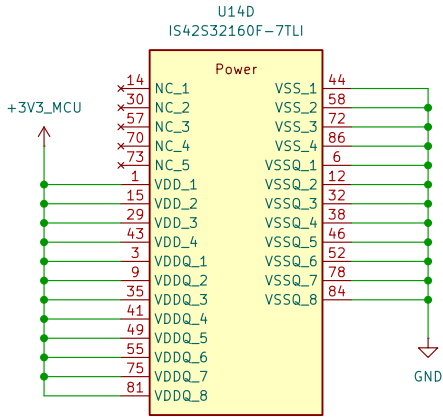
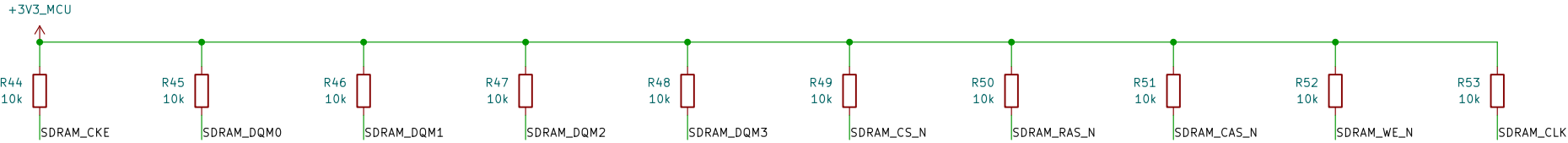
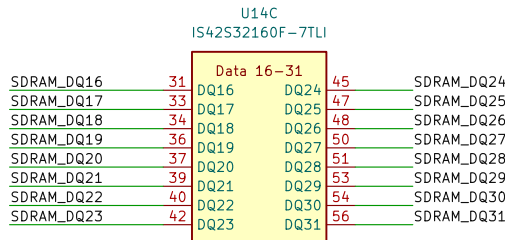
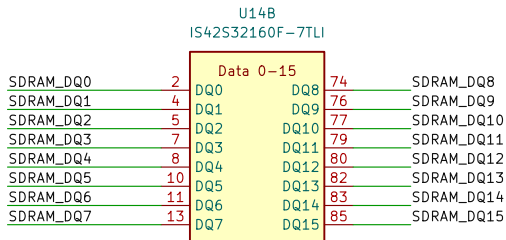
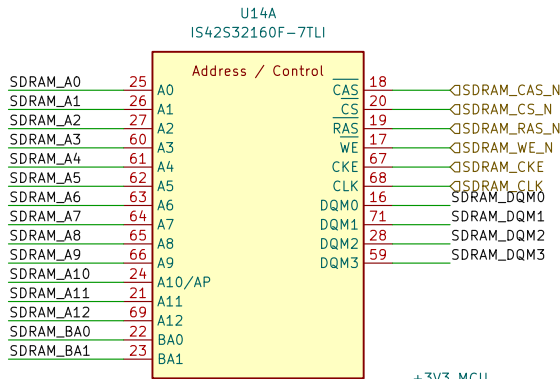
64 MiB SDRAM | ISSI IS42S32160F-7TLI  
Signal and reset—default circuits | CMS-010.  
Startup, SI/timing and hardware qualification remain open.

SDRAM\_A[0..12]

SDRAM\_BA[0..1]

SDRAM\_DQM[0..3]

SDRAM\_DQ[0..31]



CMS-010 | SDRAM power and bypass  
VDD and VDDQ share +3V3\_MCU (3.0 to 3.6 V).  
C76-C79: 4 x 100 nF for the four VDD pins.  
C80-C87: 8 x 100 nF for the eight VDDQ pins.  
C88: 10 uF local bulk; Cnom = 12 x 0.1 + 10 = 11.2 uF.  
Included in the PWR-003 1 mF whole-rail limit.  
Effective C, placement and PDN impedance require qualification.

CMS-010 | R44-R53: 10k reset defaults  
Rail 3.0..3.6V: +/-1%, 100ppm/C. |dT|<=100C.  
Rmin=10k\*0.99\*0.99=9801 ohm.  
Rmax=10k\*1.01\*1.01=10201 ohm.  
Ioff=5uA SDRAM+1uA MCU+1uA board=7uA.  
VHIGHmin=3.0-7uA\*10201=2.928593V (>2.0V VIH).  
Board leakage and temperature screen require qualification.

CMS-010 | Pull load and power states  
Sink leakage: 5uA SDRAM+1uA board=6uA.  
Isink=3.6/9801+6uA=0.373309mA (<0.374mA).  
Pmax=3.6^2/9801=1.322314mW per pull.  
All ten LOW: 10\*3.6/9801=3.673095mA rail:  
10\*3.6^2/9801=13.223140mW resistor heat.  
Reserve 4mA within the 250mA SDRAM allocation.  
CKE/DQM and CS\_N default HIGH; CLK HIGH is not a clock.  
Switched-rail pulls; self-refresh actively holds CKE LOW.  
Startup, dynamic I/O, SI and shutdown qualification OPEN.

Sheet: /64 MiB SDRAM/  
File: sdram.kicad\_sch

Title: RA8P1 E-reader - 64 MiB SDRAM

Size: A3

Date: 2026-09-08

Rev: 0.3 DRAFT

KiCad E.D.A. 10.0.5

Id: 10/10